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SET -4

Payment Processing System

App.java

```
 package com.example;


public class App {


   private double balance;



   public App(double initialBalance) {

      this.balance = initialBalance;

   }



   public String makePayment(double amount) {



      if (amount <= 0) {

         return "FAIL: Invalid amount";

      }


      if (amount > balance) {

         return "FAIL: Insufficient balance";
```

```
        }



        balance -= amount;
        return "SUCCESS";
    }



    public double getBalance() {
        return balance;
    }
}
```

AppTest.java

```
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

class AppTest {

    @Test
    void testSuccessfulPayment() {
        App app = new App(1000);

        String result = app.makePayment(500);
```

```
    assertEquals("SUCCESS", result);
    assertEquals(500, app.getBalance());
}

@Test
void testInsufficientBalance() {
    App app = new App(300);

    String result = app.makePayment(500);

    assertEquals("FAIL: Insufficient balance", result);
    assertEquals(300, app.getBalance()); // balance unchanged
}

@Test
void testInvalidAmount() {
    App app = new App(1000);

    String result = app.makePayment(-100);

    assertEquals("FAIL: Invalid amount", result);
}

@Test
void testZeroAmount() {
    App app = new App(1000);

    String result = app.makePayment(0);
```

```
        assertEquals("FAIL: Invalid amount", result);
    }
}
```

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<!-- Project Details -->
<groupId>com.example</groupId>
<artifactId>payment-system</artifactId>
<version>1.0-SNAPSHOT</version>
<!-- Java Version -->
<properties>
<maven.compiler.source>17</maven.compiler.source>
<maven.compiler.target>17</maven.compiler.target>
</properties>
<!-- Dependencies -->
<dependencies>
<!-- JUnit Dependency -->
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<version>5.9.3</version>
<scope>test</scope>
</dependency>
</dependencies>
```

```
<!--  Build Plugins  -->
<build>
<plugins>
<!--  Compiler Plugin  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.10.1</version>
<configuration>
<source>17</source>
<target>17</target>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

mvn clean test

mvn clean package

```
C:\payment-system>mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:payment-system >---------------------
[INFO] Building payment-system 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ payment-system ---
[INFO] Deleting C:\payment-system\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\payment-system\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\payment-system\target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\payment-system\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\payment-system\target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payment-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.071 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
```

```
C:\payment-system>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:payment-system >---------------------
[INFO] Building payment-system 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ payment-system ---
[INFO] Deleting C:\payment-system\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\payment-system\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\payment-system\target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\payment-system\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\payment-system\target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payment-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.078 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ payment-system ---
[INFO] Building jar: C:\payment-system\target\payment-system-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.324 s
```

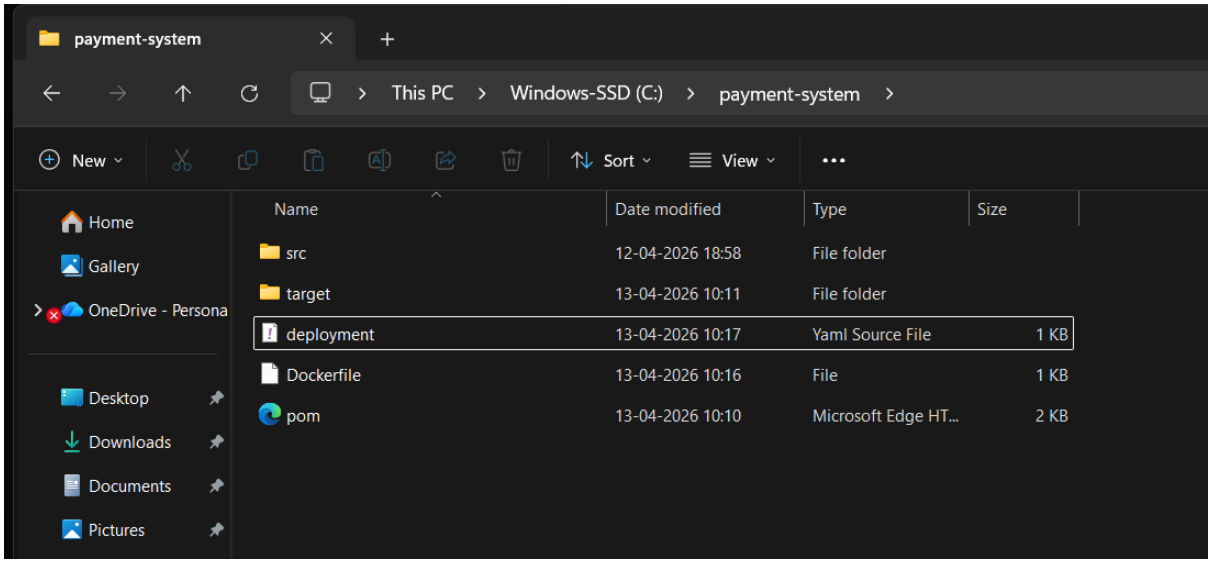
Dockerfile

```
FROM eclipse-temurin:17-jdk-alpine

WORKDIR /app

COPY target/payment-system-1.0-SNAPSHOT.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

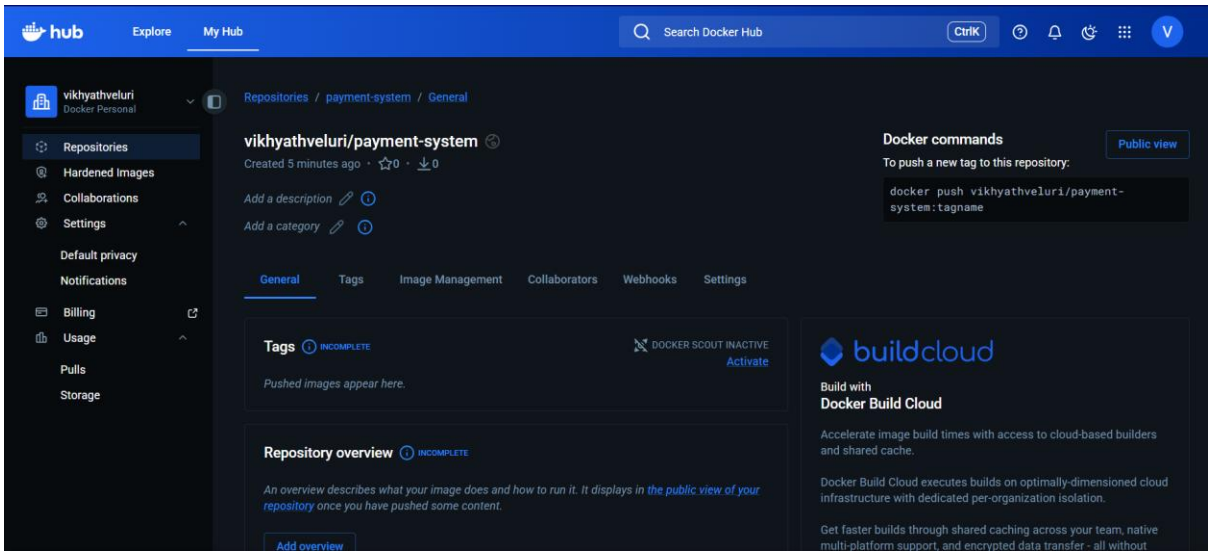
Deployment.yaml

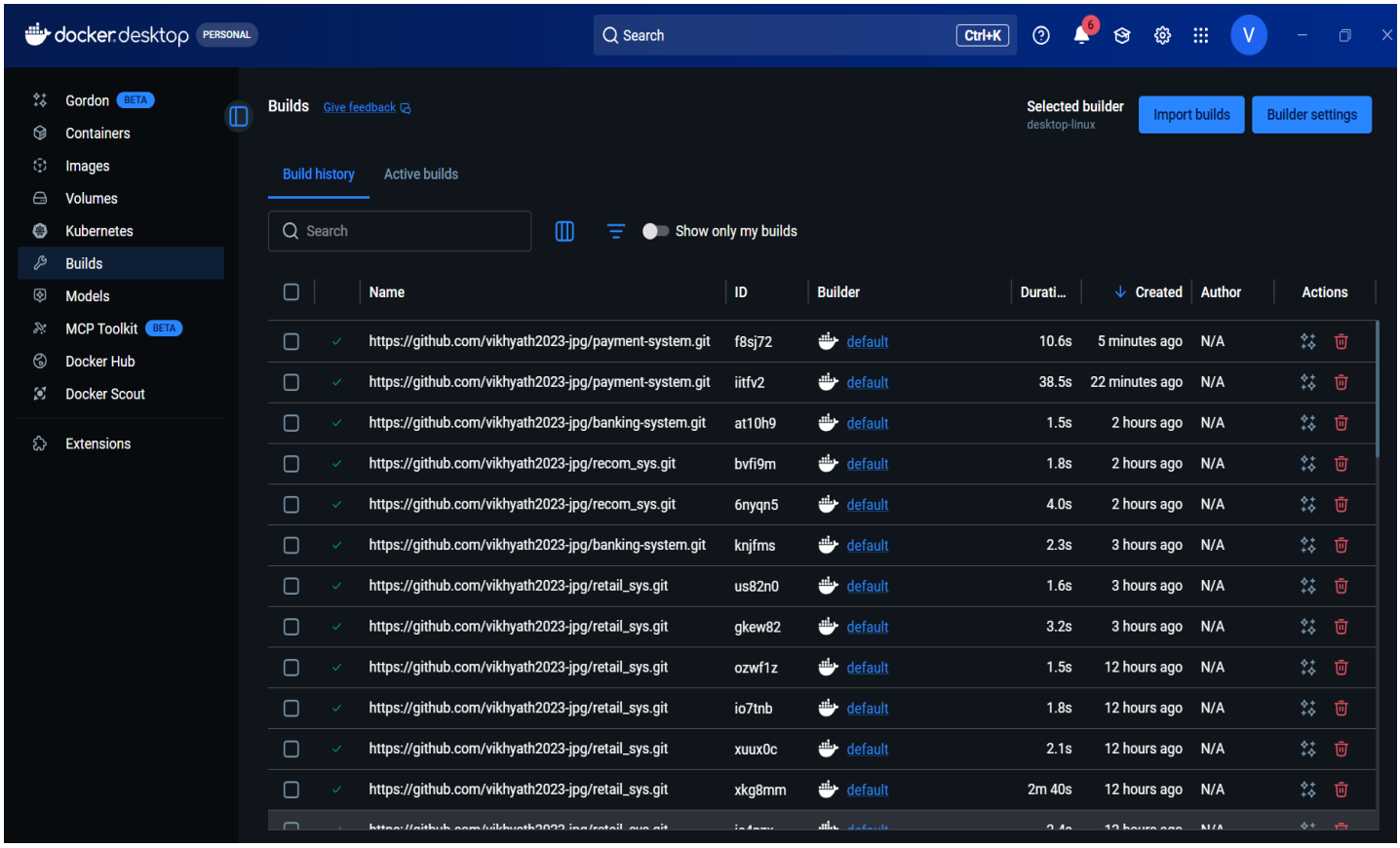
```
apiVersion: apps/v1

kind: Deployment

metadata:

  name: payment-system

  labels:

    app: java-test-app

spec:

  replicas: 2

  selector:

    matchLabels:

      app: java-test-app

  template:

    metadata:

      labels:

        app: java-test-app

    spec:

      containers:

      - name: java-test-app

        image: vikhyathveluri/payment-system:latest

        imagePullPolicy: Always

        ports:
```

```
    - containerPort: 8080


---
apiVersion: v1
kind: Service
metadata:
  name: payment-service
spec:
  type: NodePort
  selector:
    app: java-test-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
```



DockerHub

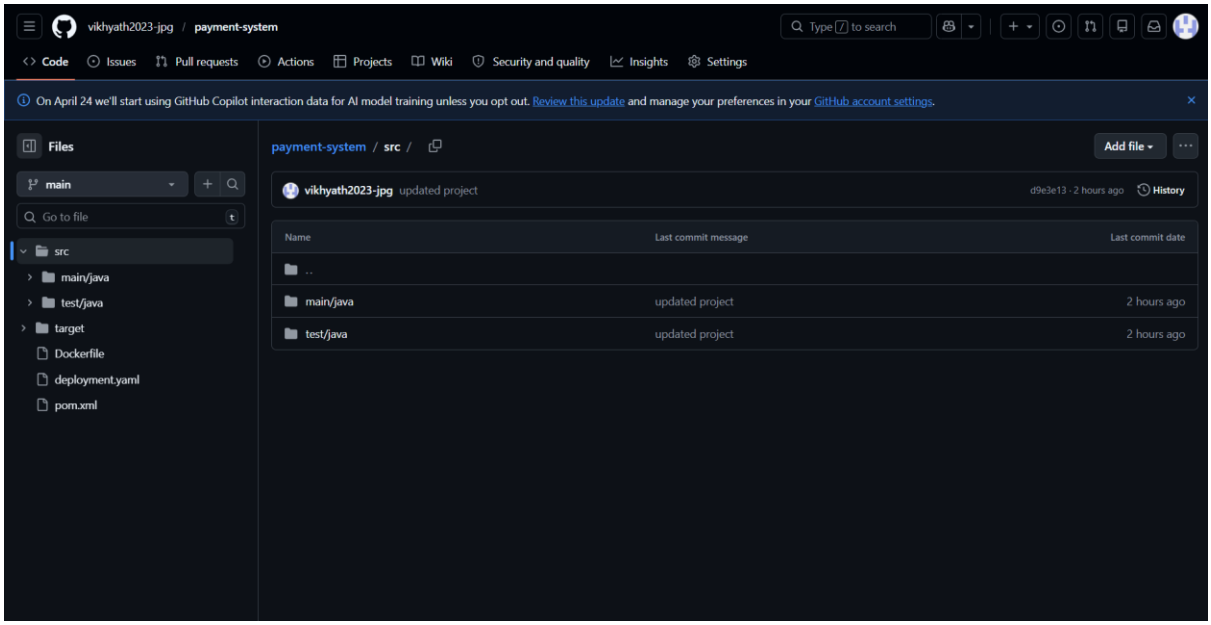


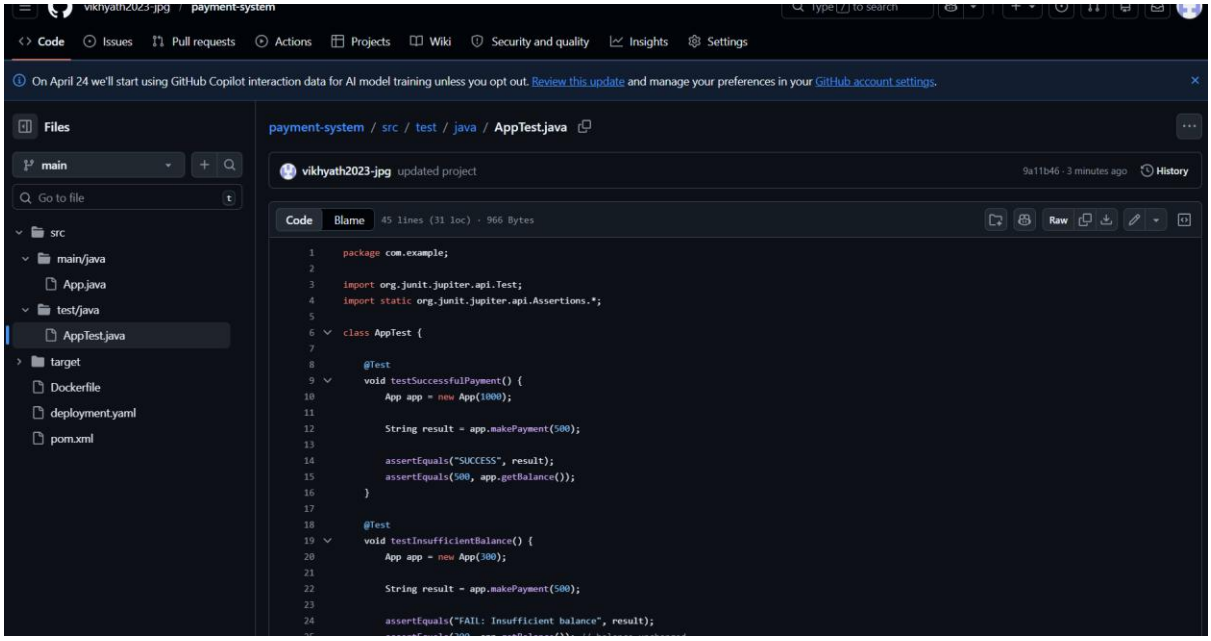


Push to Git:

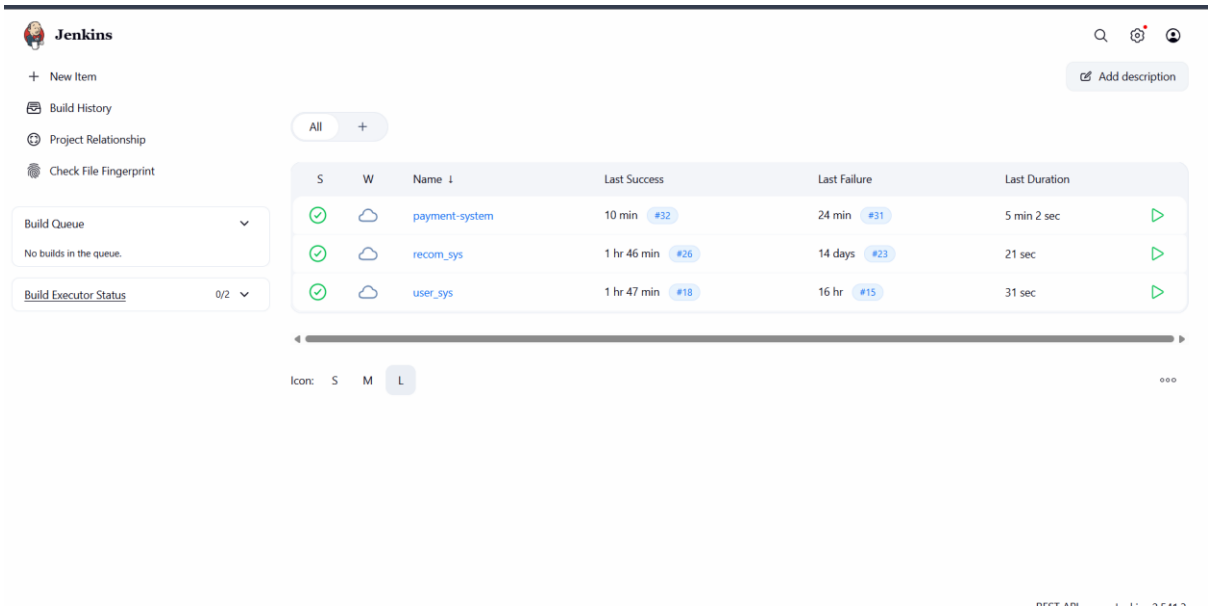
```
git init
git add .
git commit -m "Initial commit"
```

```
git branch -M main
git remote add origin https://github.com/vikhyath2023-jpg/payment-system.git
git push -u origin main
```





Jenkins:



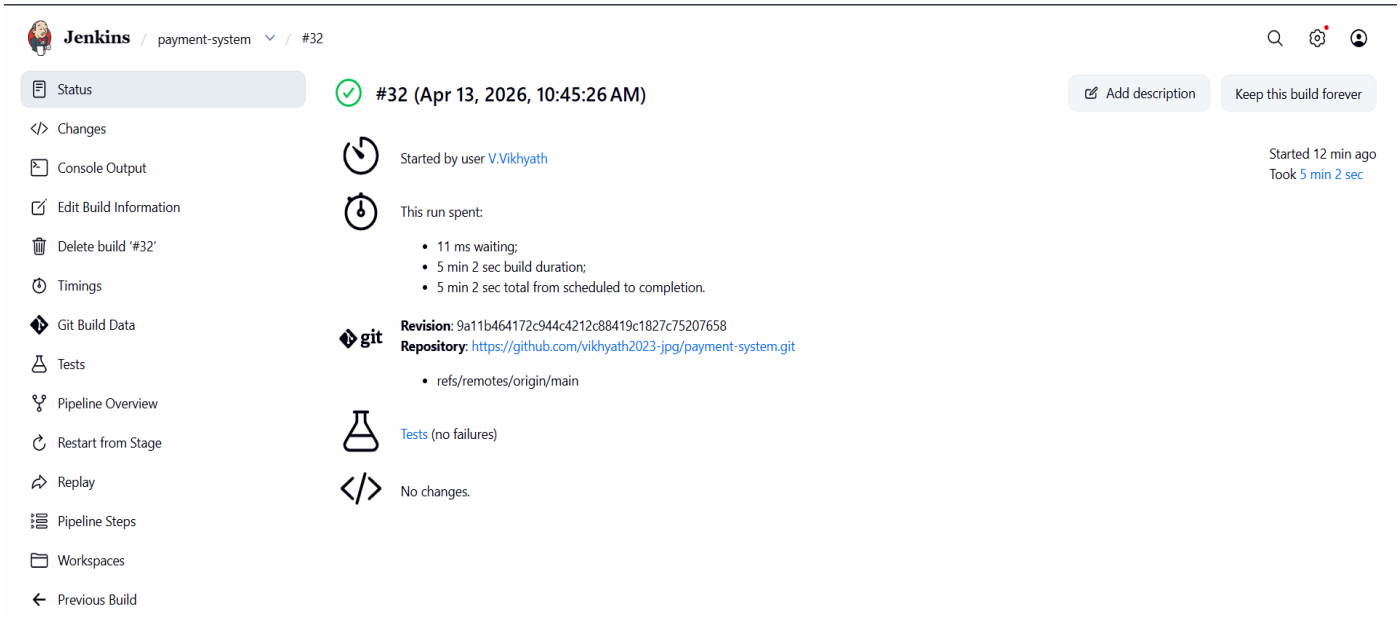
Code:

```
pipeline {
  agent any
  environment {
    DOCKER_IMAGE = "vikhyathveluri/payment-system:latest"
    KUBECONFIG = "C:\\Users\\Vikhyath\\.kube\\config"
  }
  stages {
    stage('Clone Repository') {
      steps {
        retry(2) {
```

```
        git branch: 'main',
            url: 'https://github.com/vikhyath2023-jpg/payment-system.git'
      }
    }
  }
  stage('Build & Run Tests') {
    steps {
      bat 'mvn clean test'
    }
    post {
      always {
        junit 'target/surefire-reports/*.xml'
      }
    }
  }
  stage('Package Application') {
    steps {
      bat 'mvn clean package -DskipTests'
    }
  }
  stage('Build Docker Image') {
    steps {
      bat "docker build -t %DOCKER_IMAGE% ."
    }
  }
  stage('Push to DockerHub') {
    steps {
      withCredentials([usernamePassword(
```

```
            credentialsId: 'dockerhub-creds',
            usernameVariable: 'DOCKER_USER',
            passwordVariable: 'DOCKER_PASS'
        )]) {
            bat """
            docker logout
            docker login -u %DOCKER_USER% -p %DOCKER_PASS%
            docker push %DOCKER_IMAGE%
            """
        }
    }
  }
  stage('Deploy to Kubernetes') {
    steps {
        bat """
        kubectl config use-context docker-desktop
        kubectl delete svc payment-service --ignore-not-found
        kubectl apply -f deployment.yaml
        """
    }
  }
}
post {
  success {
    echo 'Pipeline executed successfully!'
  }
  failure {
    echo 'Pipeline failed. Check logs.'
```

```
        }
    }
}
```

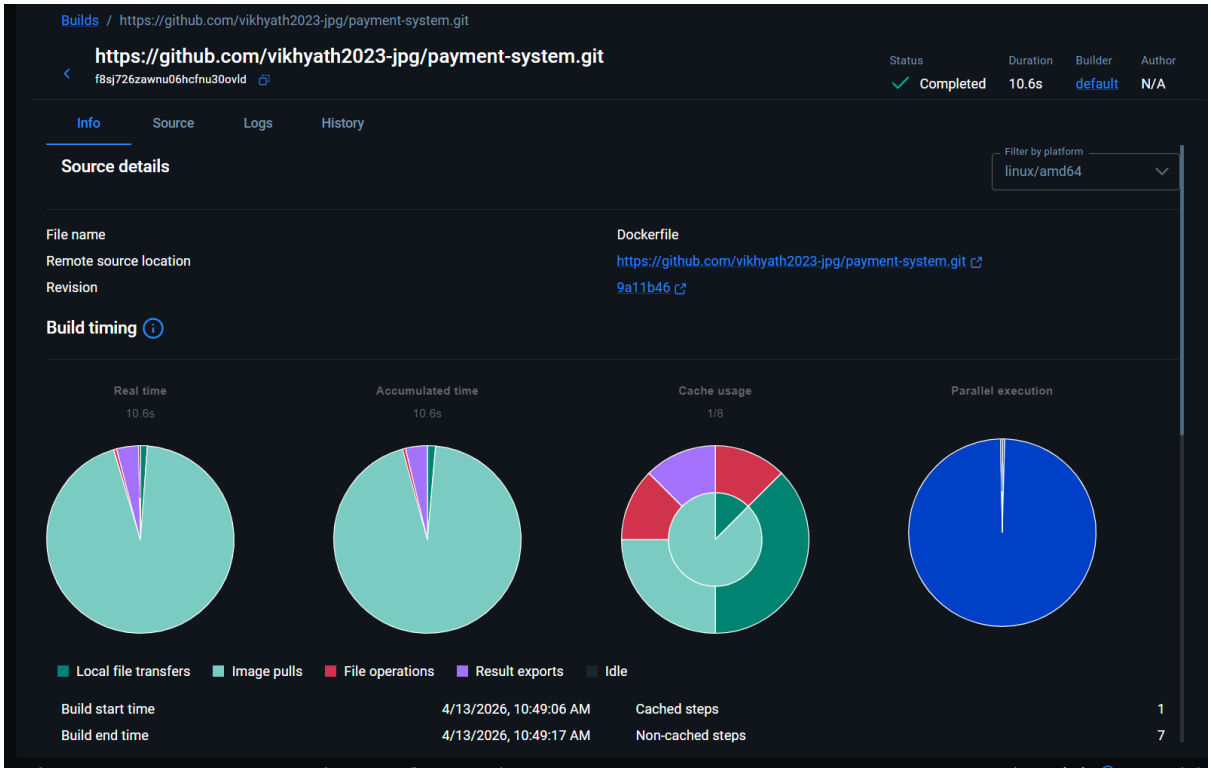


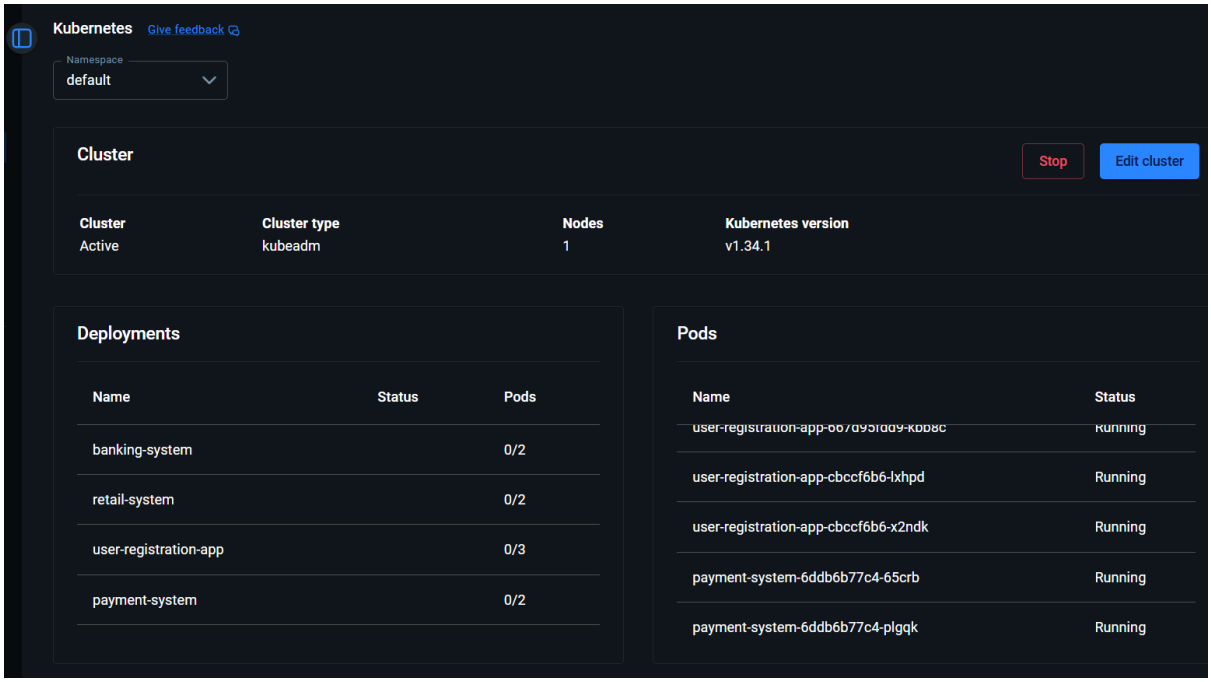
Console output:

```
[INFO] --- resources:3.4.0:testResources (default-testResources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payment-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.082 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.764 s
[INFO] Finished at: 2026-04-13T10:48:58+05:30
[INFO] ------------------------------------------------------------------------
Post stage

[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ payment-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payment-system ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ payment-system ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\payment-system\target\payment-system-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.902 s
[INFO] Finished at: 2026-04-13T10:49:05+05:30
[INFO] ------------------------------------------------------------------------
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\payment-system>kubectl config use-context docker-desktop
Switched to context "docker-desktop".

C:\ProgramData\Jenkins\.jenkins\workspace\payment-system>kubectl delete svc payment-service --ignore-not-found

C:\ProgramData\Jenkins\.jenkins\workspace\payment-system>kubectl apply -f deployment.yaml
deployment.apps/payment-system created
service/payment-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Docker Decktop:






```
C:\payment-system>kubectl get svc
NAME              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
banking-service   NodePort    10.110.146.157   <none>        80:31577/TCP   164m
kubernetes        ClusterIP   10.96.0.1        <none>        443/TCP        15d
payment-service   NodePort    10.105.45.28     <none>        80:30918/TCP   9m43s
user-service      NodePort    10.101.149.110   <none>        80:31849/TCP   111m
```